

Claire Kim

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Education

University of California, Los Angeles (UCLA) | Bachelor of Science in Mechanical Engineering **Sep 2025 – Jun 2029**

Projects

Product Engineer

July 2026 - Present

Matcha-matic | Independent Project

Camarillo, CA

- **Mechanical Design & Prototyping:** Engineered a compact, low-cost (\$50 budget) automated matcha maker from scratch in SolidWorks, utilizing FDM 3D printing and iterative design for portable on-the-go functionality
- **Mechanism & Fluid Integration:** Engineered a magnetic stirrer powertrain using a brushed DC motor, dual side magnets, and custom aerating attachments to simultaneously drive fluid mixing and micro-foam generation
- **Mechatronics & Embedded Control:** Integrated an ESP32 microcontroller, motor driver, and silicone-tube water distribution system to execute multi-mode automated programs via a 4-button user interface

Mechanical Engineering Lead

Apr 2026 - Present

Petalux | Creative Labs

Los Angeles, CA

- **Mechanical Design & Prototyping:** Engineered the mechanical structure of an automated kinetic lamp in Autodesk Fusion 360 from requirements to fully constrained CAD assemblies and prototypes
- **Mechanism & Actuation:** Designed a custom powertrain with central screw linkage and specialized petal gears to actuate a 10-petal opening sequence while minimizing axial wobble through tight tolerances
- **Cross-Functional Integration:** Developed space-efficient internal chassis routing for electronics and wiring without obstructing moving parts, while partnering with cross-functional teams to integrate user-interface hardware via snap-fits and custom cutouts

Mechanical Engineering Lead

Oct 2025 - Nov 2025

Aeris | Theta Tau

Los Angeles, CA

- **System Architecture:** Directed “Aeris”, a smart-cooling system integrating facial recognition and tri-axis actuation to achieve dynamic, automated airflow control
- **Enclosure Design & DFA:** Engineered linkages and custom enclosures in SolidWorks and Fusion 360 using Design for Assembly (DFA) principles, streamlining Arduino hardware integration via high-precision friction fits
- **Prototyping & Validation:** Executed the full product lifecycle from FDM 3D printing to electromechanical validation, performing precision soldering and iterative troubleshooting to eliminate hardware errors

Chassis Mechanical Engineer

Sept 2025 - June 2026

BB-8 | X1 Robotics - American Society of Mechanical Engineers

Los Angeles, CA

- **Mechanism Design:** Designed and assembled BB-8 drive mechanisms in SolidWorks, optimizing weight distribution and omnidirectional motion stability
- **Electromechanical Packaging:** Positioned motor housings and drivetrain components to fit within a strict 2mm machined shell tolerance while maximizing internal volume
- **Structural Analysis:** Modeled the core support ring in CAD, validating structural stress tolerances and in-house manufacturing feasibility for rapid fabrication

Experience

RCMakes

Mar 2022 - Jul 2025

Lead Intern & Jobshop Lead

Camarillo, CA

- **Digital Fabrication & Mentorship:** Managed 80+ interns in digital fabrication workflows, providing technical instruction on CNC machining, laser cutting, and 3D printing to generate \$5k in total shop revenue
- **Technical Operations & Outreach:** Developed an Arduino robotics curriculum for STEM workshops and directed advanced theater technology, including lighting, sound, and projection, for 500-seat community events

Technical Skills

- Modeling: SolidWorks, Fusion360, FEA, AutoCAD, DFM, Tolerance Analysis, Rapid Prototyping
- Electronics & Manufacturing: Arduino, Soldering, Machining, Welding, 3D Printing, CNC
- Programming Languages: Python, C++, JavaScript, CSS/HTML
- Relevant Coursework: Statics, Engineering Design/CAD, MATLAB Programming